

Python Unit 3 – Question Bank

Complex Data Types (Strings, Lists, Tuples, Dictionaries, Functions)

Section A: Theory Questions

Strings

- 1 What is a string in Python? List the three ways of creating a string with examples.
- 2 What is an empty string? How is it different from a string containing only spaces?
- 3 Explain why strings are called immutable in Python. What error occurs if you try to modify a character of a string directly?
- 4 Differentiate between forward indexing and backward indexing in strings with a suitable example.
- 5 What does "traversing a string" mean? List the three ways to traverse a string in Python.
- 6 Differentiate between `find()` and `index()` methods of a string.
- 7 What is the difference between `str1 == str2` and `str1 is str2`?
- 8 Explain string slicing syntax `string[start:end]`. What happens when start or end is omitted?
- 9 What is the use of the `ord()` and `chr()` functions? Give one example each.
- 10 How does Python compare two strings internally? Explain with an example using ASCII values.
- 11 Differentiate between `isalpha()` and `isalnum()` string methods.
- 12 What is the difference between `split()` and `join()` methods?

Lists

- 13 What is a list in Python? List any four key features of a list.
- 14 Differentiate between a list and a tuple.
- 15 What is a nested list? Give an example and explain how to access its elements.
- 16 Differentiate between `del`, `pop()`, and `remove()` for deleting elements from a list.
- 17 What is the difference between `append()` and `extend()` methods of a list?
- 18 Explain the concept of list slicing with the syntax `L[start:stop]`.
- 19 What does the `sort()` method do? How is it different from using `reverse=True` as a parameter?
- 20 Can a list contain elements of different data types? Justify with an example.

Tuples

- 21 What is a tuple? How is it different from a list?
- 22 Why do we write `(5,)` instead of `(5)` to create a single-element tuple?
- 23 Name the only two built-in methods available for tuples and explain why tuples support just these two.
- 24 List the three main operations that can be performed on tuples.
- 25 Why is `cmp()` not available in Python 3? What alternative do we use instead?

Dictionaries

- 26 What is a dictionary in Python? How is it different from a list in terms of structure and ordering?
- 27 Explain the difference between `del dict[key]` and `dict.clear()`.
- 28 What is the use of the `get()` method? How is it better than directly accessing a key using `dict[key]`?
- 29 Explain the difference between `keys()`, `values()`, and `items()` methods with examples.
- 30 How does the `update()` method work when merging two dictionaries? What happens if both dictionaries have a common key?

Functions

- 31 What is a function? Why are functions used in programming?
- 32 Differentiate between a parameter and an argument.
- 33 What are the three categories of functions in Python? Explain each briefly.
- 34 What are default arguments? What rule must be followed while defining them?
- 35 Explain the flow of execution when one function calls another function, with a suitable example.
- 36 Can a function in Python return more than one value? Discuss briefly.

Section B: Coding Questions

Strings

- 1 Write a program to check whether a given string is a palindrome or not.
- 2 Write a program to count the number of vowels and consonants in a string entered by the user.
- 3 Write a program to reverse a string without using the built-in `reversed()` function.
- 4 Write a program that takes a sentence as input and counts the number of words in it (using `split()`).
- 5 Write a program to find the ASCII value of each character of a string and print them.
- 6 Write a program to check whether two strings are anagrams of each other.
- 7 Write a program to replace all occurrences of a character in a string with another character (without using `replace()`).
- 8 Write a program to count how many times a particular substring occurs in a string.
- 9 Write a program to convert the first letter of every word in a sentence to uppercase (without using `title()`).

Lists

- 10 Write a program to find the largest and smallest number in a list without using `max()` or `min()`.
- 11 Write a program to remove duplicate elements from a list.
- 12 Write a program to merge two lists and sort the resulting list in ascending order.
- 13 Write a program that accepts a list of numbers and separates them into two lists: even numbers and odd numbers.

- 14 Write a program to find the sum and average of elements in a list.
- 15 Write a program to insert an element at a specific position in a list without using insert().
- 16 Write a program to count the occurrences of each element in a list and display them.
- 17 Write a program to reverse a list without using the reverse() method.

Tuples

- 18 Write a program to swap two numbers using a tuple.
- 19 Write a program to find the maximum and minimum value in a tuple without using max() and min().
- 20 Write a program that converts a list into a tuple and vice versa.
- 21 Write a program to count how many times a given value appears in a tuple using the count() method.
- 22 Write a program to concatenate two tuples and then slice the first four elements from the result.

Dictionaries

- 23 Write a program to count the frequency of each character in a string using a dictionary.
- 24 Write a program to merge two dictionaries. If a key exists in both, keep the value from the second dictionary.
- 25 Write a program to find the key with the maximum value in a dictionary.
- 26 Write a program that accepts student names and marks as key-value pairs, and prints the name of the topper.
- 27 Write a program to remove a key from a dictionary only if it exists (avoid raising an error).
- 28 Write a program to sort a dictionary by its values in descending order.

Functions

- 29 Write a function is_prime(n) that returns True if n is a prime number, otherwise False.
- 30 Write a function factorial(n) using default arguments, where n defaults to 5 if not provided.
- 31 Write a function calculate(a, b, operation) that performs addition, subtraction, multiplication, or division based on the value of operation.
- 32 Write a function that takes a list as an argument and returns a new list containing only the elements greater than a given threshold.
- 33 Write a function word_lengths(sentence) that returns a dictionary mapping each word in the sentence to its length.
- 34 Write a function to check whether a given number is an Armstrong number.

Section C: Mixed / Application-Based Questions (Higher Order)

- 1 Write a program that takes a paragraph as input, and using strings, lists, and dictionaries together, prints the 3 most frequently occurring words.
- 2 Write a program to implement a simple contact book using a dictionary, where each key is a name and the value is a tuple of (phone number, email).

- 3 Write a program that reads a list of tuples (name, marks) and returns a dictionary sorted by marks in descending order.
- 4 Write a function-based program to check if a given string is a valid palindrome after removing all non-alphanumeric characters and ignoring case.
- 5 Write a program using functions to convert a list of Celsius temperatures into Fahrenheit, and separately return the maximum and minimum Fahrenheit values.

Question bank based on Unit 3: Python Complex Data Types — Strings, Lists, Tuples, Dictionaries, and Functions.